EliteDesign: Elite-Guided Learning for RNA Design via the Cross-Entropy Method

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Abstract

The goal of RNA design is to find, within an exponentially large sequence space, a nucleotide sequence that folds into a specified target structure. While essential for developing RNA-based therapeutics such as riboswitches and ribozymes, efficiently identifying sequences with high target folding probability remains challenging, especially for long sequences. Currently, the state-of-the-art method, SamplingDesign, suffers from extreme computational inefficiency. To address this bottleneck, we present EliteDesign, a novel framework that leverages the cross-entropy method to iteratively refine a distribution over candidate RNA sequences. On the Eterna100 benchmark, EliteDesign achieves a higher average target folding probability than SamplingDesign while running 500× faster. Furthermore, EliteDesign scales efficiently to long RNA structures far beyond the reach of SamplingDesign: on ten 16S rRNA targets (950–1,525 nucleotides), it achieves an average target folding probability of 0.51 versus 0.20 for the strongest scalable baseline, SAMFEO, while accelerating runtime by 24×.